

## Article

# A Miniaturized Ultrawideband Frequency-Selective Rasorber with High Absorptivity

Jiayao Luo <sup>1</sup>, Hao Wen <sup>1</sup>, Liping Yan <sup>1,2,\*</sup> , Xiang Zhao <sup>1</sup>  and Changjun Liu <sup>1,2</sup> 

<sup>1</sup> College of Electronics and Information Engineering, Sichuan University, Chengdu 610064, China; luo2023222055118@stu.scu.edu.cn (J.L.); 2024222055018@stu.scu.edu.cn (H.W.); zhaoxiang@scu.edu.cn (X.Z.); cjliu@scu.edu.cn (C.L.)

<sup>2</sup> Yibin Industrial Technology Research Institute, Sichuan University, Yibin 644000, China

\* Correspondence: liping\_yan@scu.edu.cn

## Abstract

To overcome the intrinsic trade-off among miniaturization, ultrawideband (UWB) performance, and structural simplicity in conventional frequency-selective rasorber (FSR) design, this paper proposes a miniaturized UWB absorption–transmission–absorption (A-T-A) FSR based on an inter-cell current-interaction mechanism. The structure comprises a dielectric matching layer (DML), a lossy frequency-selective surface (FSS), a lossless FSS layer, and air/dielectric spacers. Both FSS layers are fabricated on Rogers 4350B substrates without any metallized via or multiple lossy/lossless FSS stacking. The proposed FSR achieves a miniaturized structure with dimensions of  $0.085 \lambda_L \times 0.085 \lambda_L \times 0.118 \lambda_L$  (where  $\lambda_L$  corresponds to the wavelength at the lowest absorption frequency). A fractional operational bandwidth around 144% is obtained, covering 2.88–12.87 GHz and 14.98–17.61 GHz with absorptivity over 80%, together with a low-loss transmission band of 13.57–14.56 GHz exhibiting a minimum insertion loss of 0.41 dB. As the incident angle increases up to 40°, the FSR retains more than 134% bandwidth for both TE and TM polarizations. A prototype was fabricated and measured, and the results agree well with the simulations.

**Keywords:** frequency-selective rasorber (FSR); miniaturized structure; wideband absorption performance

## 1. Introduction

A frequency-selective rasorber (FSR) is a periodic electromagnetic structure that integrates frequency-selective transmission and broadband absorption properties, allowing electromagnetic waves to pass within a designated band while absorbing out-of-band radiation. This unique property makes FSRs highly attractive for stealth radomes, electromagnetic protection, and compatibility enhancement [1,2]. FSRs can be classified from both structural and functional perspectives. Structurally, they include 2-D [3], 2.5-D [4] and 3-D [5] configurations. Functionally, they can be categorized into low-frequency transmission/high-frequency absorption (T-A) [6,7], low-frequency absorption/high-frequency transmission (A-T) [8–10], and absorption on both sides of the passband (A-T-A) types [11–13], respectively. Compared to the T-A and A-T types, the A-T-A type FSR offers superior stealth and enhanced electromagnetic interference (EMI) shielding capability.

Despite extensive research, the design of high-performance A-T-A FSRs still faces a fundamental and unresolved contradiction. Miniaturization severely restricts the achievable inductance (L) and capacitance (C), while ultrawideband A-T-A behavior requires



Academic Editor: Zaharias Zaharis

Received: 14 February 2026

Revised: 17 March 2026

Accepted: 18 March 2026

Published: 24 March 2026

**Copyright:** © 2026 by the authors.

Licensee MDPI, Basel, Switzerland.

This article is an open access article distributed under the terms and conditions of the [Creative Commons Attribution \(CC BY\)](https://creativecommons.org/licenses/by/4.0/) license.

multiple well-distributed resonances. Miniaturized 2-D layouts enable dense periodic packing [10–13], and thereby mitigate edge diffraction effects and facilitate conformal deployment on curved surfaces. However, the reduced electrical size limits the design space for generating the equivalent L and C required to excite multiple resonances, which is essential for wideband absorption [14–16]. As a result, most miniaturized FSRs suffer from insufficient bandwidth. Further, another inherent challenge is the trade-off between strong out-of-band absorption and low in-band insertion loss (IL).

The ideal FSR for practical applications should simultaneously satisfy often-conflicting requirements: a compact unit cell, ultrawide absorption bandwidth, high absorptivity, low in-band IL, polarization insensitivity and angular stability. In pursuing these objectives—specifically balancing ultrawideband performance with structural miniaturization—researchers have developed various design strategies. Key approaches involve utilizing 2.5-D configurations to enhance inductance [2,4,16,17], loading lumped elements to refine frequency response [14,18–20], and employing compact resonators to strengthen coupling [21–26]. However, these existing approaches still suffer from limitations, including: (i) insufficient achievable bandwidth [22]; (ii) degradation of polarization stability during miniaturization [5]; and (iii) increased structural complexity due to multilayer FSSs, vias, or excessive lumped loading [12], which raises fabrication costs and reduces reliability.

To address these challenges, this paper introduces a fundamentally different design method. Instead of relying on localized resonators within isolated unit cells, adjacent cells are interconnected through strip lines to form a unit-interaction grating topology. This configuration significantly extends the effective current paths and enhances inter-cell coupling, enabling distributed resonances across multiple cells. As a result, multiple absorption poles and a transmission pole are realized within a miniaturized footprint, without requiring vias, multilayer metallic stacks, or complex lumped loading. Both simulations and measurements verify that this mechanism enables ultrawideband A-T-A performance while maintaining low in-band loss and excellent polarization/angle stability.

## 2. Design and Analysis of FSR Structure

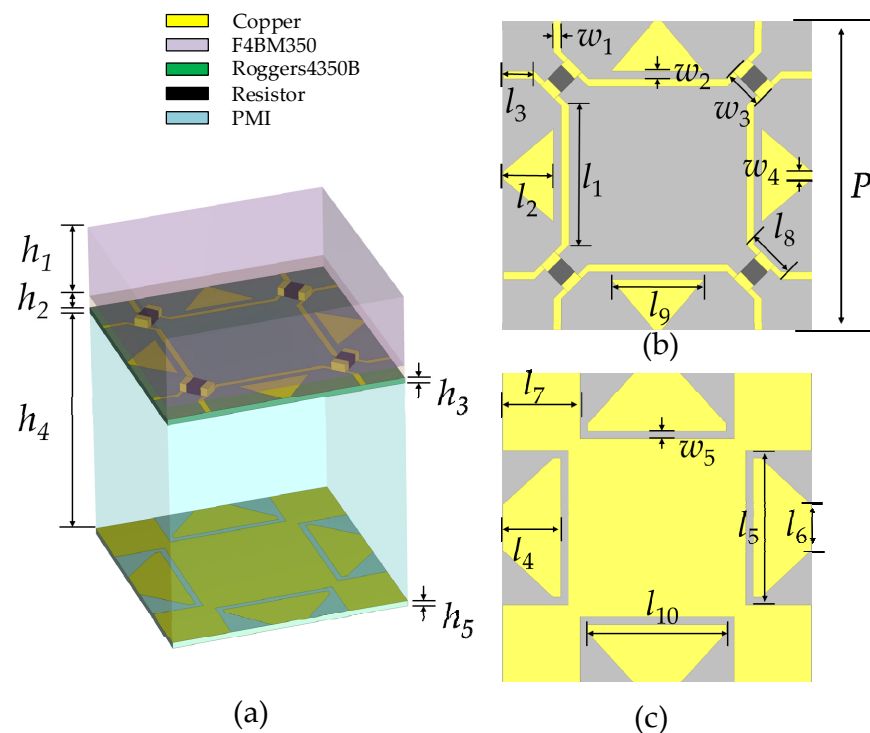
### 2.1. Configuration of the FSR Unit Cell

Figure 1 illustrates the geometry of the proposed ultrawideband FSR, which features a miniaturized unit cell. The FSR structure consists of five layers: a top dielectric matching layer (DML), a thin air spacer, a lossy FSS, a dielectric spacer made of polymethacrylimide (PMI) foam ( $\epsilon_r = 1.05$ ), and a bottom lossless FSS.

The lossy FSS layer is composed of four metallic half-rectangular loops with two notched corners. Each half-loop is connected to its counterpart in the adjacent unit cell, forming a complete loop across the periodic array. The long edges of the half-loops are interconnected through four lumped resistors, constituting a resistively loaded square loop. To introduce a transmission pole within the absorption band, a trapezoidal metallic patch is embedded inside each half-loop, as shown in Figure 1b.

The lossless FSS layer is designed to reflect incident waves within the absorption bands while allowing transmissions at the passband. This function is realized by etching a semi-annular slot formed by an outer rectangular contour and an inner trapezoidal contour on each side of the metallic ground plane, as shown in Figure 1c.

Both FSS layers are fabricated on a Rogers 4350B substrate (relative permittivity  $\epsilon_r = 3.66$ , loss tangent  $\tan \delta = 0.004$ , thickness = 0.2 mm) without metallic vias. A 2.5 mm thick DML made of F4BM350 is placed above the lossy layer to improve impedance matching with free space, thereby broadening the absorption bandwidth. A 0.43 mm thick air spacer is introduced to accommodate the thickness of the lumped resistors.



**Figure 1.** Configuration of the proposed FSR: (a) 3D perspective view, and top views of (b) lossy and (c) lossless layers.

The unit cell is arranged in a  $90^\circ$  rotational symmetry, ensuring polarization-insensitive behavior. The optimized geometrical parameters are listed in Table 1.

**Table 1.** Optimized geometrical parameters of the proposed FSR unit cell.

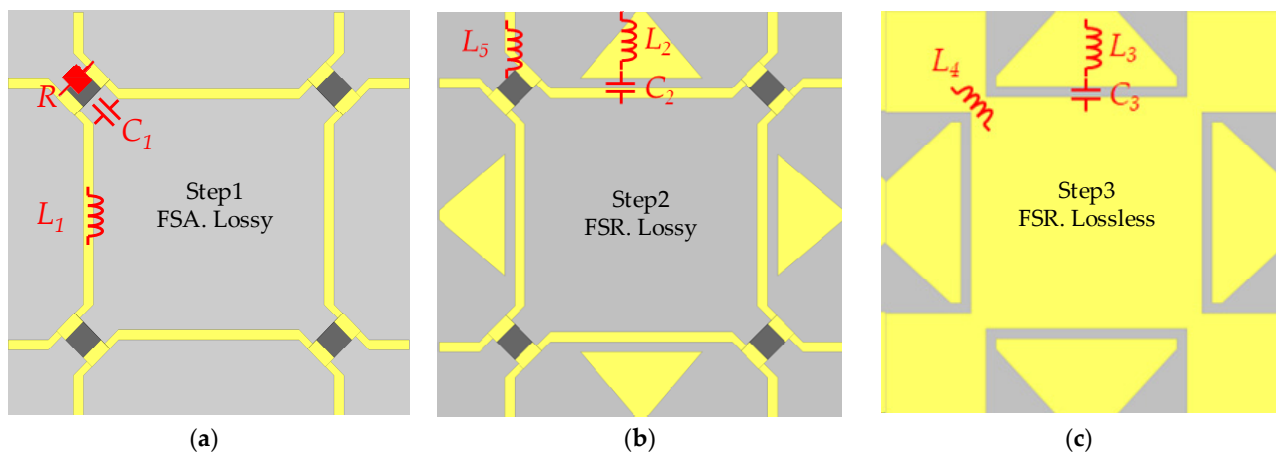
| Variable | Value | Variable | Value | Variable | Value | Variable | Value |
|----------|-------|----------|-------|----------|-------|----------|-------|
| $h_1$    | 2.5   | $w_1$    | 0.2   | $l_1$    | 3.7   | $l_6$    | 1.2   |
| $h_2$    | 0.43  | $w_2$    | 0.2   | $l_2$    | 1.3   | $l_7$    | 2.0   |
| $h_3$    | 0.2   | $w_3$    | 1.0   | $l_3$    | 0.8   | $l_8$    | 1.27  |
| $h_4$    | 7.8   | $w_4$    | 0.2   | $l_4$    | 1.5   | $l_9$    | 2.4   |
| $h_5$    | 0.2   | $w_5$    | 0.2   | $l_5$    | 4.0   | $l_{10}$ | 3.7   |

All the values are in mm.

## 2.2. Design Principle of FSR

The design evolution of the lossy layer is illustrated in Figure 2. The process begins with a loop-type FSS absorber (FSA) formed by resistively loading a square loop composed of four half-rectangular loops, as shown in Figure 2a. Unlike conventional isolated resonators in each unit cell, the loops here are connected across adjacent cells. This inter-cell current sharing mechanism enables current paths with multiple effective lengths, thereby naturally producing multiple resonances required for broad absorption.

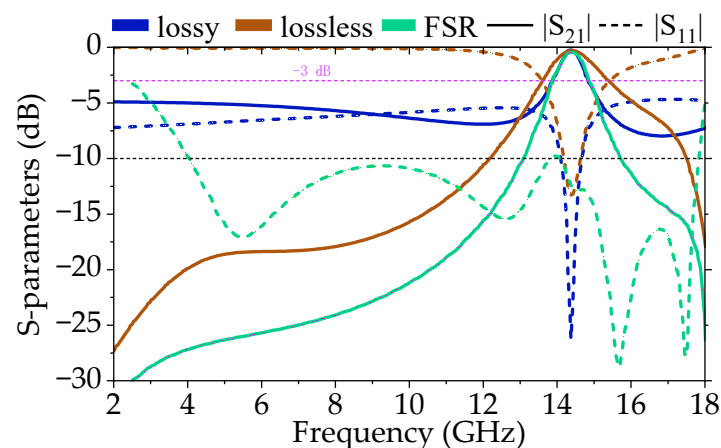
To create a transmission window within the absorption band, trapezoidal patches are inserted into the half-rectangular loops, forming a distributed series-parallel LCL resonant path, as depicted in Figure 2b. The resonant frequency for the transmission band is  $f_{tp} = 1/2\pi\sqrt{(L_2 + L_5)C_2}$ . To achieve a steep roll-off around the passband, a relatively large capacitance  $C_2$  is required [3]. A similar LCL resonance is implemented in the lossless FSS layer through slot etching. With the passband center frequency fixed, the trapezoid dimensions are tuned to obtain the desired  $C_2$  and  $L_2$ . The parameters of the lossless FSS layer are then optimized so that its transmission band fully overlaps that of the lossy layer.



**Figure 2.** Design evolution of the FSR with corresponding equivalent lumped elements: (a) lossy layer of the FSA, (b) lossy layer of the FSR, and (c) lossless layer of the FSR.

A critical design consideration for minimizing in-band insertion loss (IL) is the placement of the trapezoidal patches at the inter-cell junctions. This position effectively decouples the passband resonant current from the resistor-loaded branches, suppressing ohmic dissipation and preserving transmission efficiency.

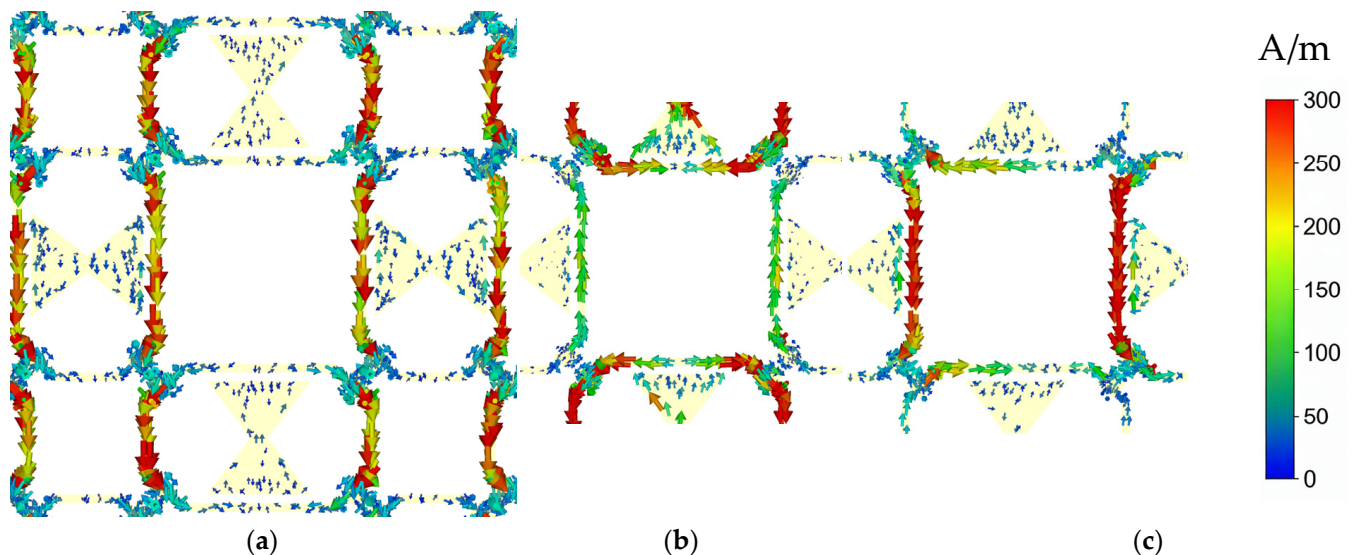
To validate the design principle, Figure 3 illustrates the S-parameters of the separated lossless layer and lossy FSS layers, as well as the FSR. Note that the electromagnetic response results from strong coupling among the lossy FSS layer, the lossless FSS layer, and the dielectric substrates. Therefore, the lossy and lossless FSS layers are analyzed separately while retaining the corresponding dielectric substrates.



**Figure 3.** Simulated S-parameters of the lossless FSS, lossy FSS, and proposed FSR.

As shown in Figure 3, the lossless layer exhibits in-band transmission and out-of-band reflection, whereas the lossy layer demonstrates in-band transmission and out-of-band absorption. Since the lossless layer is absent in the isolated lossy-layer simulation, the reflected field required to establish strong absorption is insufficient, resulting in relatively weak absorptivity. However, the overall FSR response indicates excellent in-band transmission with an insertion loss (IL) of 0.41 dB and out-of-band absorption with a 90% absorptivity bandwidth of 134.2%.

To clarify the operating mechanism, the surface current density distributions on the lossy layer under TE polarization at representative resonant frequencies are analyzed, as shown in Figure 4.



**Figure 4.** Surface current density distributions at (a) 7.5 GHz, (b) 14 GHz, and (c) 17.2 GHz.

At the lower absorption band, currents predominantly flow along the resistor-connected metallic strips with in-phase distributions across adjacent cells. For example, the resonance around 7.5 GHz involves current spreading across three unit cells, as shown in Figure 4a. As frequency decreases, the effective current path adaptively extends across more unit cells to match the longer resonant wavelength. This behavior clearly demonstrates the strong inter-cell coupling mechanism, where current is guided through the lumped resistors and efficiently dissipated via ohmic loss. At higher frequencies (e.g., 17.2 GHz), the resonance becomes localized within a single unit cell, with strong current confined to the central square loop and dissipated by the resistors at corners, as shown in Figure 4c.

In contrast, at the passband around 14 GHz, the dominant current flows through the trapezoidal patch and the surrounding rectangular loop, as shown in Figure 4b, while currents through the resistor-loaded branches remain minimal, leading to low insertion loss.

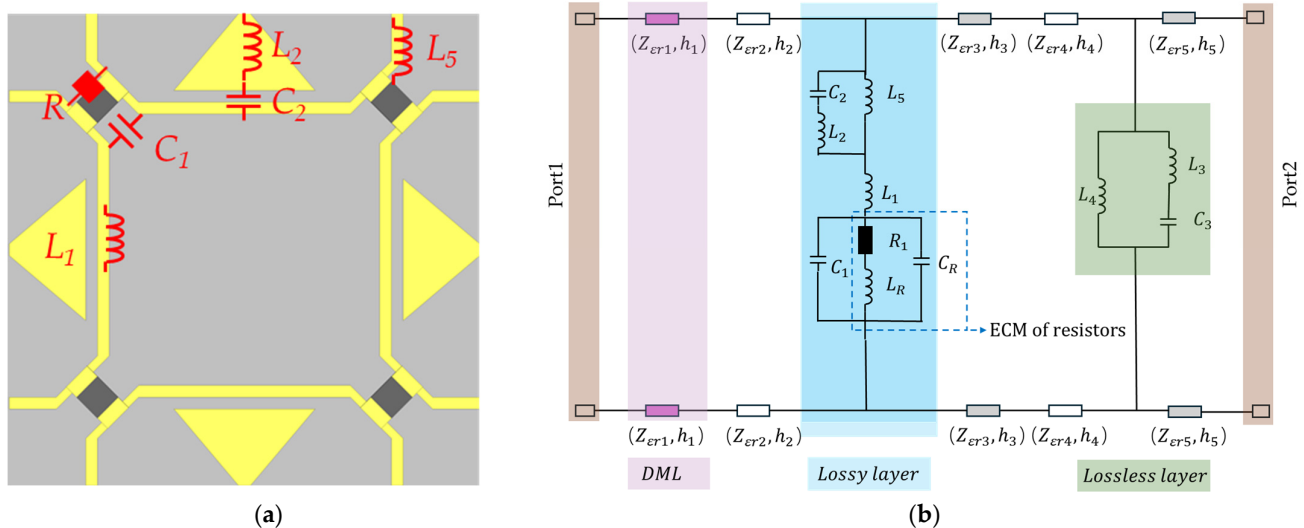
### 2.3. Equivalent Circuit Model

The operating principle of the FSR can also be interpreted using the equivalent circuit model (ECM) shown in Figure 5. The mapping between the lossy FSS metallic pattern and the lumped elements is illustrated in Figure 5a, while the mapping for lossless FSS can be referred to in Figure 2, both for the case of TE polarization.

The metallic strips in the lossy FSS correspond to inductors  $L_1$ ,  $L_2$ , and  $L_5$ , respectively. The gaps used for resistor soldering are represented by capacitor  $C_1$ , while the gap between the trapezoidal patch and the half-rectangular loop is modeled as capacitor  $C_2$ . Similarly, in the lossless layer, the metallic patches are modeled as inductors  $L_3$  and  $L_4$ , while the gap is modeled by capacitor  $C_3$ . The lumped-element values of the ECM are summarized in Table 2.

In addition, the parasitic effects of the resistors in a 0402 package are considered by including inductance  $L_R$  and capacitance  $C_R$  in the resistor equivalent circuit model. The values of  $L_R$  and  $C_R$  are 0.906 nH and 0.037 pF, respectively [3].

All the dielectric substrates and spacers are modeled as transmission lines with a characteristic impedance of  $Z = Z_0 / \sqrt{\epsilon_{ri}}$  and a length of  $h_i$ , where  $Z_0 = 377 \Omega$ ,  $\epsilon_{r1} = 3.50$ ,  $\epsilon_{r2} = 1$ ,  $\epsilon_{r3} = \epsilon_{r5} = 3.66$ , and  $\epsilon_{r4} = 1.05$ ; the values of  $h_i$  are provided in Table 1.

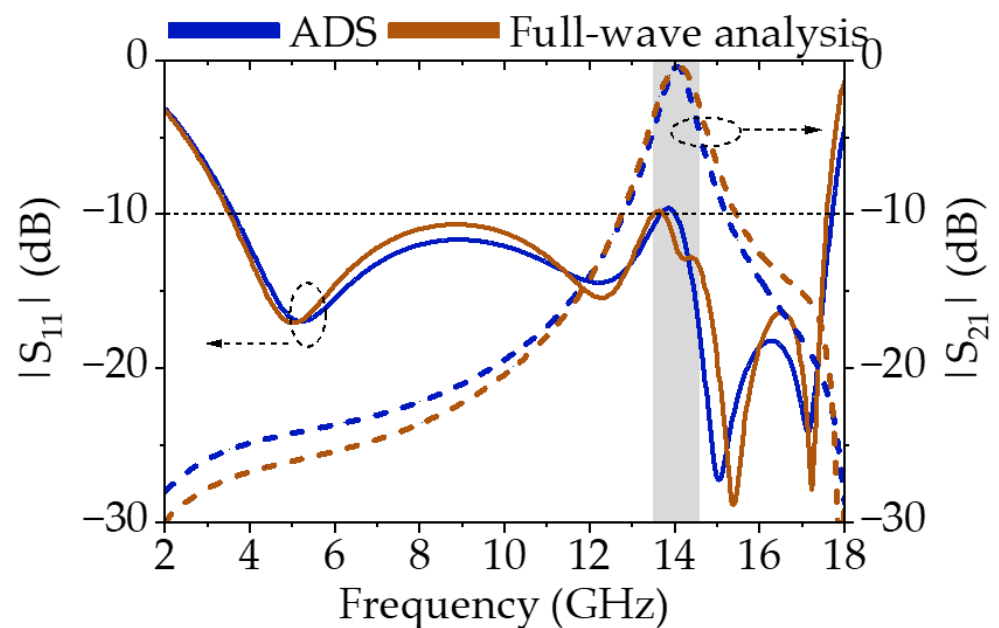


**Figure 5.** (a) Mapping between the lossy FSS metallic pattern and the lumped elements. (b) Equivalent circuit model of the proposed FSR.

**Table 2.** Optimized values of the ECM variables.

| Variable | Value        | Variable | Value    | Variable  | Value        |
|----------|--------------|----------|----------|-----------|--------------|
| $C_1$    | 0.022 pF     | $L_1$    | 1.74 nH  | $Z_{er1}$ | 201 $\Omega$ |
| $C_2$    | 0.125 pF     | $L_2$    | 0.125 nH | $Z_{er2}$ | 369 $\Omega$ |
| $C_3$    | 0.059 pF     | $L_3$    | 1.3 nH   | $Z_{er3}$ | 197 $\Omega$ |
| $C_R$    | 0.037 pF     | $L_4$    | 0.707 nH | $Z_{er4}$ | 369 $\Omega$ |
| $R_1$    | 250 $\Omega$ | $L_5$    | 0.34 nH  | $Z_{er5}$ | 197 $\Omega$ |
|          |              | $L_R$    | 0.906 nH |           |              |

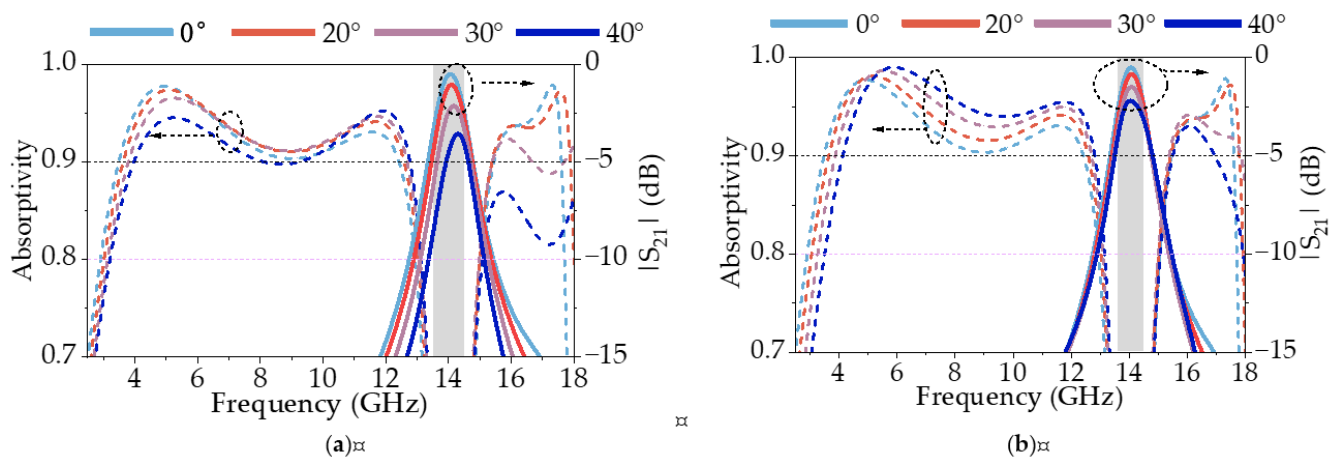
The S-parameters predicted by the ECM are compared with full-wave simulations using CST, as shown in Figure 6. The good agreements observed validate the ECM analysis.



**Figure 6.** Comparison of S-parameters between the ECM and full-wave simulations.

#### 2.4. Performance of the Proposed FSR

The absorptivity and transmission characteristics under normal and oblique incidence for both TE and TM polarizations are simulated using CST, as depicted in Figure 7.



**Figure 7.** Simulation of the FSR performance for (a) TE and (b) TM polarization.

Under normal incidence, the passband ( $|S_{21}| > -3$  dB) spans 13.57–14.56 GHz with a minimum insertion loss (IL) of 0.41 dB at 14.0 GHz. The lower and upper absorption bands with absorptivity over 80% cover 2.88 GHz to 12.87 GHz (LAB) and 14.98 GHz to 17.61 GHz (UAB), corresponding to fractional bandwidths of 127.0% and 16.1%, respectively, where absorptivity is calculated as  $1 - |S_{11}|^2 - |S_{21}|^2$ . The sharp transition between absorption and transmission bands indicates the relatively high frequency selectivity of the FSR.

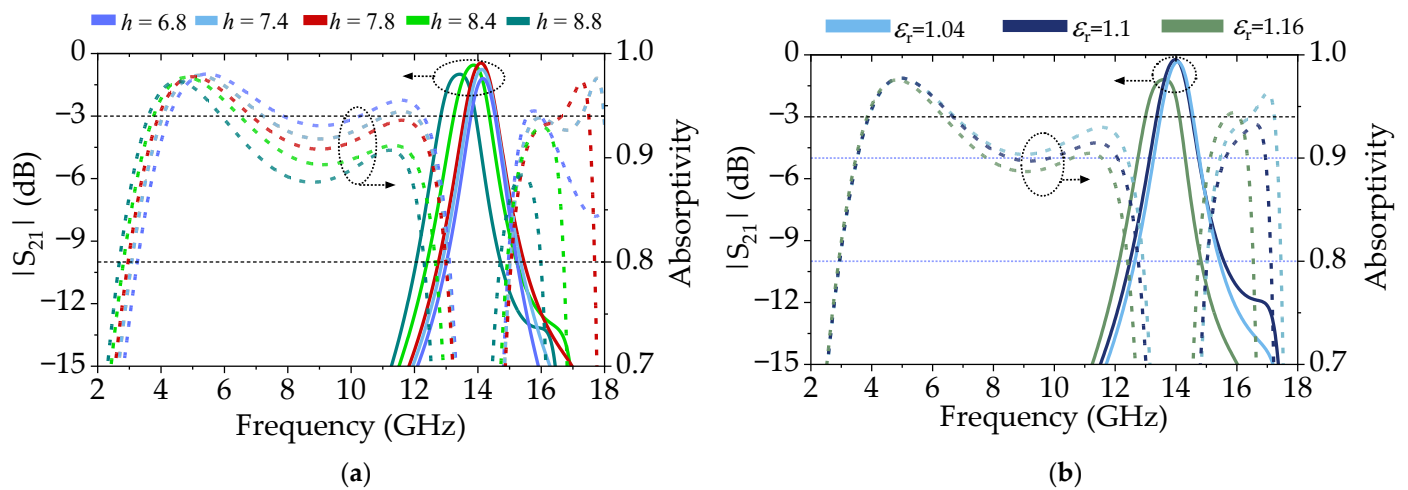
As the incident angle increases to 40°, the 80% absorptivity bandwidths still remain 121.5% (LAB) and 19.2% (UAB) for TE polarization, with a minimum IL of 3.54 dB at 14.28 GHz. For TM polarization, the absorption bandwidths remain 115.5% (LAB) and 17.2% (UAB), with a minimum IL of 2.17 dB at 14.08 GHz. These results confirm excellent polarization insensitivity and angular stability.

Considering fabrication tolerance and assembly errors, a comprehensive parametric study was carried out to examine the influences of the key configuration parameters, spacer thickness  $h_4$ , relative permittivity of PMI, and chip resistor value, on the FSR performance.

Figure 8a shows the effects of the spacer thickness  $h_4$  on the FSR performance. The variation range of 6.8 mm to 8.8 mm is chosen based on possible assembly tolerances. When  $h_4 = 7.8$  mm, the passband is centered at 14.0 GHz with an IL of 0.41 dB. As  $h_4$  increases, both the LAB and UAB shift toward lower frequencies, accompanied by a noticeable decrease in absorptivity and bandwidth, particularly for the UAB. The transmission passband also shifts toward lower frequencies, reaching 13.43 GHz, with IL increasing to 0.97 dB when  $h_4 = 8.8$  mm. When  $h_4$  decreases, both the LAB and UAB shift toward higher frequencies, and the 80% absorptivity bandwidth is slightly enhanced. The passband center shifts a little bit to 14.14 GHz with an IL reduction to 1.17 dB at  $h_4 = 6.8$  mm. Importantly, the 80% absorptivity bandwidth remains stable when  $h_4$  varies within the considered tolerance range, maintaining a fractional bandwidth of approximately 123.0%. These results indicate that the proposed FSR exhibits good tolerance to spacer thickness variations, which is beneficial for practical fabrication and assembly.

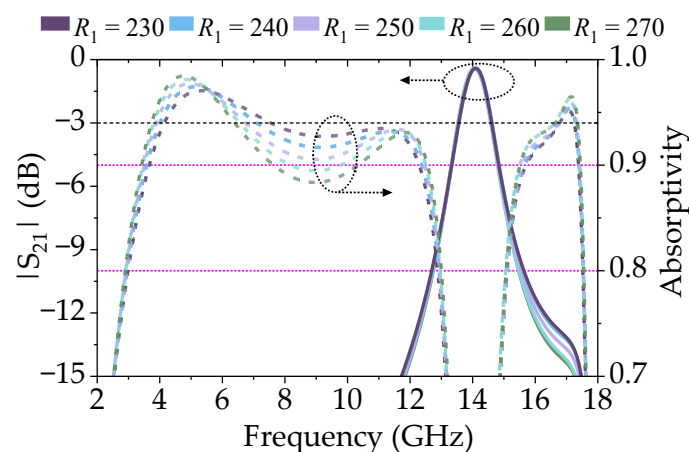
As the relative permittivity increases, the LAB upper bound shifts slightly downward from 12.91 GHz to 12.4 GHz, and the UAB shifts from 15.02–17.45 GHz to 14.74–16.54 GHz. For the transmission window, the passband moves from 13.5–14.5 GHz to 13.01–14.12 GHz, with an IL increase from 0.41 dB to 1.19 dB. Despite these variations, the 80% absorptivity

band still remains 134.5%. These results shown in Figure 8b indicate that the proposed FSR design is relatively robust to realistic variations in the spacer permittivity.



**Figure 8.** Effect of PMI interlayer properties on absorption and transmission: (a) thickness and (b) permittivity.

The chip resistors integrated into the FSR serve primarily to dissipate incident electromagnetic energy. As aforementioned, the passband resonant elements are strategically designed to decouple from the resistor-loaded branches. Consequently, the resistance fluctuation exerts a negligible impact on the passband, as observed in Figure 9, where the passband frequency and insertion loss remain almost unchanged. However, the resistance value has a more noticeable effect on the absorption level, since it directly controls the power dissipation in the lossy layer. Specifically, moderate variations in resistance produce only minor shifts in the 80% absorptivity bandwidth. However, the 90% absorptivity bandwidth becomes more sensitive, particularly in the LAB. When the resistance increases from  $250 \Omega$  to  $270 \Omega$ , the LAB region with absorptivity above 90% becomes narrower, while the UAB experiences a slight bandwidth reduction. Overall, the results indicate that the proposed design maintains stable broadband absorption performance even under realistic resistor tolerances.

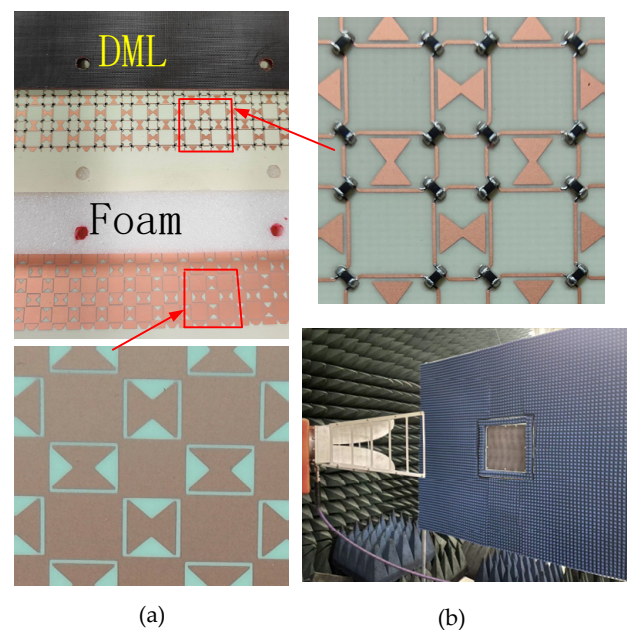


**Figure 9.** Effect of the loaded resistance on absorption and transmission.

### 3. Experimental Validation

#### 3.1. Fabrication and Test Setup

To experimentally validate the electromagnetic characteristics of the proposed FSR, a prototype consisting of a  $25 \times 25$  unit-cell array was fabricated, as shown in Figure 10a. All layers of the FSR prototype were mechanically assembled using nylon screws and nuts to avoid undesired metallic perturbations. The S-parameters were measured using a free-space measurement setup. An Agilent E8363C PNA network analyzer (10 MHz–40 GHz), together with a pair of broadband horn antennas operating over 2–18 GHz, was employed to measure the transmission coefficient  $|S_{21}|$  and reflection coefficient  $|S_{11}|$ , as illustrated in Figure 10b.



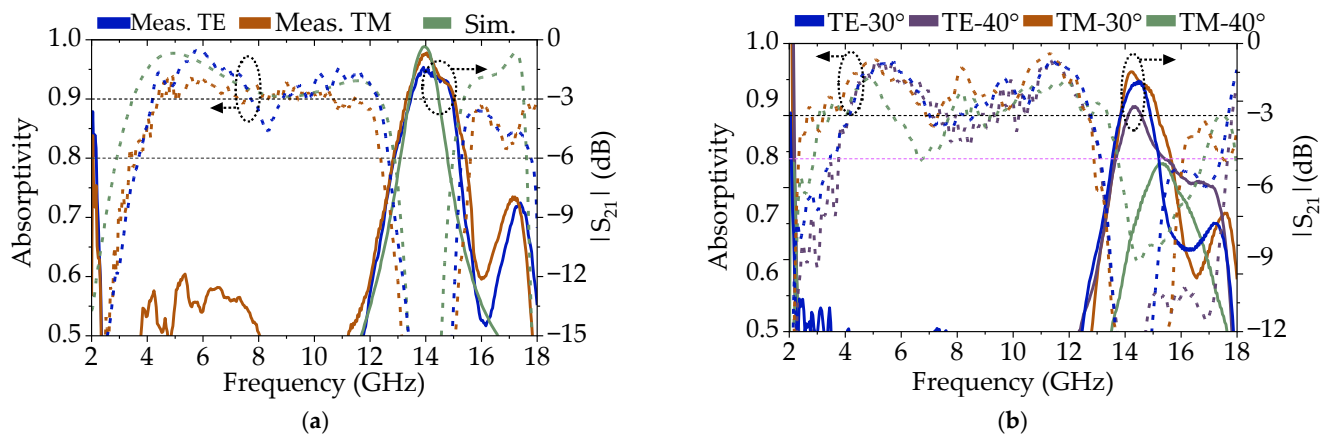
**Figure 10.** Photographs of the FSR prototype (a) and experimental setup (b).

#### 3.2. Measurement Results

Figure 11a illustrates the measured S-parameters under normal incidence, in comparison with simulated results. A clear A-T-A response is observed, and good agreement between measurements and simulations is obtained for both TE and TM polarizations. For normal incidence, the operating frequency bands range from 3.68 GHz to 17.76 GHz and from 3.36 GHz to 18 GHz for TE and TM polarizations, respectively. For TE polarization, the measured transmission window ( $|S_{21}| > -3$  dB) ranges from 13.48 to 14.92 GHz, with a minimum IL of 1.39 dB at 13.92 GHz. For TM polarization, the transmission band spans 13.32–14.98 GHz, with a minimum IL of 0.71 dB at 13.96 GHz.

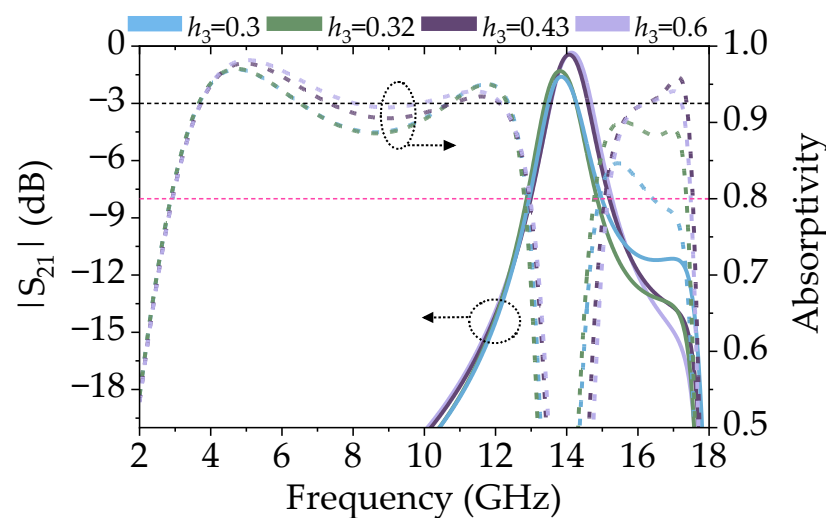
Figure 11b presents the measured S-parameters under oblique incidence. At  $30^\circ$  incidence, the FSR still maintains robust performance with strong out-of-band absorption and a minimum IL of 1.62 dB (TE) and 1.18 dB (TM). When the incident angle further increases to  $40^\circ$ , the transmission band and the upper absorption band (UAB) of the TM polarization begin to degrade, and the minimum insertion loss exceeds 3 dB.

The discrepancies between the measurement and simulation are primarily attributed to: (i) fabrication tolerances and assembly misalignment, (ii) resistance variation of chip resistors, and (iii) diffraction due to the limited size of the sample, particularly at low frequencies.



**Figure 11.** Measured results of the proposed FSR (a) under normal incidence and (b) under oblique incidence.

Notably, fabrication tolerances and assembly error can cause the spacer thickness  $h_3$  to deviate from its nominal value of 0.43 mm. Figure 12 evaluates this effect by simulating the FSR performance with varying  $h_3$ . The results indicate that varying  $h_3$  greatly influences IL and the absorptivity performance at high frequencies, with absorptivity even lower than 80% when  $h_3$  decreases to 0.3 mm. Therefore, this assembly error also contributes to the deterioration of measured results.



**Figure 12.** Influence of spacer thickness  $h_3$  on the FSR performance.

#### 4. Comparison with Related Studies

The performance of the proposed FSR is compared with representative state-of-the-art A-T-A FSRs in the literature, as summarized in Table 3. Refs. [18,19] achieve excellent bandwidth and miniaturization. However, both designs employ two to three times the number of lumped elements and additional lossy/lossless FSS layers, thereby at the cost of increased design complexity, structural intricacy and fabrication expense. Compared to FSRs with four resistors per unit cell, the proposed design employs only a single lossy FSS layer and a single lossless FSS layer, achieving superior comprehensive performance in terms of ultrawideband operation, miniaturization, high absorption, and low insertion loss.

**Table 3.** Performance comparison to recent related works.

| Ref.      | Num.-LE | Unit Size ( $\lambda_L$ ) | Thickness ( $\lambda_L$ ) | Num. FSS Layers      | PB (GHz)/IL (dB)            | FBW (80%) | FBW (90%) | LAB (GHz)           | UAB (GHz)           | AS    |
|-----------|---------|---------------------------|---------------------------|----------------------|-----------------------------|-----------|-----------|---------------------|---------------------|-------|
| [2]—2021  | 4       | 0.26                      | 0.32                      | 2 lossy + 2 lossless | 7.37–13.13/(0.05)           | 123%      | * 113%    | 3.67–6.80 (58.2%)   | 13.74–15.31 (10.7%) | 30/30 |
| [3]—2025  | 4       | 0.1                       | 0.1                       | 1 lossy + 1 lossless | 10.04–10.56/(0.48)          | 139%      | * 132%    | 3.61–9.82 (92.5%)   | 11.03–20.22 (58.8%) | 55/55 |
| [11]—2024 | 4       | 0.14                      | 0.12                      | 1 lossy + 2 lossless | 10.10–13.01/(0.14)          | 135%      | \         | 3.69–8.54 (79.3%)   | 14.18–18.96 (28.8%) | 30/30 |
| [12]—2022 | 4       | 0.154                     | 0.077                     | 1 lossy + 1 lossless | 9.75–10.34/(0.25)           | 113%      | \         | 3.84–9.21 (82.3%)   | 10.91–13.9 (24.1%)  | 40/40 |
| [15]—2022 | 12      | 0.062                     | 0.06                      | 2 lossy + 1 lossless | 3.81–4.44, 6.28–8.49/(0.82) | 124%      | * 118%    | 1.98–2.76 (32%)     | 6.28–8.49 (30%)     | 40/40 |
| [17]—2022 | 4       | 0.112                     | 0.113                     | 1 lossy + 3 lossless | 6.67–10.71 (1 dB)/(0.1)     | 124%      | * 113%    | 3.7–5.1 (31.84%)    | 8.2–15.8 (63.51%)   | 45/45 |
| [25]—2023 | 4       | 0.15                      | 0.12                      | 1 lossy + 1 lossless | 7.66–11.76 (1 dB)/(0.4)     | 136%      | \         | 2.93–6.57 (81.6%)   | 12.52–15.43 (20.8%) | 30/30 |
| [26]—2023 | 4       | 0.079                     | 0.078                     | 1 lossy + 1 lossless | * 4.95–5.5/(0.46)           | 131%      | \         | 2.14–4.54 (71.8%)   | 5.94–10.32 (53.9%)  | 40/40 |
| [18]—2026 | 12      | 0.075                     | 0.12                      | 2 lossy + 3 lossless | 7.9–12.1 (1 dB)/(0.13)      | * >164%   | 164%      | 1.81–6.7 (115%)     | 13–18.55 (35%)      | 30/30 |
| [19]—2026 | 8       | 0.064                     | 0.11                      | 2 lossy + 2 lossless | 8.0–10.5 (0.38)             | 171%      | \         | 1.60–6.64 (122.3%)  | 11.62–20.78 (56.5%) | 30/30 |
| This work | 4       | 0.085                     | 0.118                     | 1 lossy + 1 lossless | 13.57–14.56 (0.41)          | 144.0%    | 134.2%    | 2.88–12.87 (127.0%) | 14.98–17.61 (16.1%) | 40/40 |

Num.-LE: number of lumped elements. PB: passband. LAB and UAB: lower and upper absorption bandwidth with absorptivity above 80%. AS: angular stability. \* values estimated from the graphical results reported in the literature.

Overall, the proposed FSR structure avoids multilayer lossless FSSs, metallic vias, or excessive lumped-element loading, resulting in a simpler, more robust, and fabrication-friendly architecture. It also demonstrates superior integrated absorption–transmission performance compared with most existing designs, making it particularly attractive for compact absorptive radome and electromagnetic stealth applications.

## 5. Conclusions

A novel miniaturized, ultrawideband FSR is presented. By exploiting a unit-interaction design in the lossy FSS layer, extended current paths and distributed resonances are realized, enabling ultrawideband A-T-A performance within a simple 2-D, via-free structure. The proposed FSR achieves a transmission band centered at 14.0 GHz with a minimum insertion loss of 0.41 dB. The absorptivity above 80% is obtained over 2.88–12.87 GHz (127.0%) and 14.98–17.61 GHz (16.1%), while maintaining stable performance up to 40° oblique incidence for both TE and TM polarizations. Experimental measurements closely match the simulation results, confirming the validity of the proposed design and its underlying physical mechanism.

The miniaturized unit cell of the proposed design makes it particularly suitable for conformal absorber applications and deployment in space-constrained environments, as it can better accommodate increased surface curvature. In addition, the proposed FSR architecture can potentially be extended to reconfigurable designs through the integration of active components such as PIN diodes or varactors. Owing to the grid-type configuration, all adjacent unit cells are interconnected, enabling unified bias control without the need for an additional biasing network. Nevertheless, the incorporation of active components inevitably introduces additional design complexity, increased insertion loss, and fabrication challenges. Furthermore, practical deployment in conformal radome environments may

require careful consideration of the mechanical robustness and environmental stability of multilayer dielectric substrates.

**Author Contributions:** Conceptualization, J.L. and L.Y.; Methodology, J.L. and H.W.; Writing—original draft preparation, J.L. and H.W.; Writing—review and editing, L.Y. and C.L.; Visualization, J.L. and X.Z.; Supervision, L.Y.; Funding acquisition, L.Y. and C.L. All authors have read and agreed to the published version of the manuscript.

**Funding:** This research was funded in part by the National Natural Science Foundation of China (NSFC), grant number U22A2015, and in part by the Sichuan Science and Technology Program, grant number 2024YFHZ0282.

**Data Availability Statement:** The original contributions presented in this study are included in the article. Further inquiries can be directed to the corresponding author.

**Conflicts of Interest:** The authors declare no conflicts of interest.

## References

1. Jia, Y.; Zhai, H.; Guo, C.; Song, L. A Dual-Band Composite Frequency Selective Rasorber with Broadband Absorption Performance. *IEEE Antennas Wirel. Propag. Lett.* **2023**, *22*, 1992–1996. [\[CrossRef\]](#)
2. Yang, Z.; Jiang, W.; Huang, Q.; Hong, T. A 2.5-D Miniaturized Frequency-Selective Rasorber with a Wide High-Transmission Passband. *IEEE Antennas Wirel. Propag. Lett.* **2021**, *20*, 1140–1144. [\[CrossRef\]](#)
3. Ai, J.; Hong, J.; Gao, R.X.K.; Liu, C.; Zhao, X.; Yan, L. Enhanced Equivalent Circuit Model-Aided Broadband Single-Lossy-Single-Lossless-Layer Rasorber Design with High Selectivity. *IEEE Antennas Wirel. Propag. Lett.* **2025**, *24*, 1080–1084. [\[CrossRef\]](#)
4. Yu, Q.; Liu, S.; Monorchio, A.; Kong, X.; Brizi, D.; Wu, C.; Wen, Y. A Highly Selective Rasorber with Ultraminiaturized Unit Based on Interdigitated 2.5-D Parallel Resonator. *IEEE Trans. Electromagn. Compat.* **2022**, *64*, 1585–1592. [\[CrossRef\]](#)
5. Wang, Y.; Qi, S.S.; Shen, Z.; Wu, W. Ultrathin 3-D Frequency Selective Rasorber with Wide Absorption Bands. *IEEE Trans. Antennas Propag.* **2020**, *68*, 4697–4705. [\[CrossRef\]](#)
6. Yu, Y.; Zhang, C.; Liu, Q.; Liao, Z.; Peng, L. High-Selectivity Band-Absorptive Frequency-Selective Rasorber. *IEEE Antennas Wirel. Propag. Lett.* **2024**, *23*, 2623–2627. [\[CrossRef\]](#)
7. Hu, X.J.; Shen, Z.; Wu, D.L. Bandpass Frequency-Selective Surface with Stable Transmission Amplitude and Insertion Phase Delay. *IEEE Trans. Antennas Propag.* **2025**, *73*, 5384–5392. [\[CrossRef\]](#)
8. Zhu, S.; Wang, Y.; Wang, Z.; He, J.; Quan, X.; Gao, D.; Cao, Z. On the Equivalent-Circuit-Based Design of Double-Lossy-Layer Wide Transmissive Rasorbers with Ultrawide Reflection Suppression Band. *IEEE Trans. Antennas Propag.* **2024**, *72*, 4999–5010. [\[CrossRef\]](#)
9. Dai, Y.; Fu, Y.; Chen, S.; Wu, J.; Liu, S.; Shen, Z.; Yang, H. Miniaturized frequency selective rasorber with absorption in S–C band and transmission in X band. *Phys. Scr.* **2023**, *98*, 025506. [\[CrossRef\]](#)
10. Wang, B.; Ding, X.; Cai, Y.; Wu, Y.; Li, W.; Luo, Z.; Si, K.; Cao, Z.; Jiang, J.; Miao, L. A Broadband Dual-Polarization Transmission Frequency Selective Rasorber with Low Insertion Loss. *IEEE Antennas Wirel. Propag. Lett.* **2025**, *24*, 2754–2758. [\[CrossRef\]](#)
11. Wu, K.; Xu, W.; Yang, Y.; Li, P. A Dual-Polarized C-X-Ku-Band Frequency-Selective Rasorber with Wide High-Transmission Passband. *IEEE Trans. Electromagn. Compat.* **2024**, *66*, 1295–1299. [\[CrossRef\]](#)
12. Shi, R.; Yu, S.; Kou, N. A Miniaturized Frequency-Selective Rasorber with Absorption Bands on Two Sides of Passband for Antenna Dome. *IEEE Antennas Wirel. Propag. Lett.* **2022**, *21*, 2161–2165. [\[CrossRef\]](#)
13. Xing, Z.; Cui, J.; Yang, P.; Wang, Q.; Yang, G.; Li, J.; Wang, L. Broadband Transmission Rasorber with Dual-Broadband Absorption by High-Accuracy Equivalent Admittance Analysis Method with Convergent Solution Conditions. *IEEE Trans. Electromagn. Compat.* **2025**, *67*, 1423–1431. [\[CrossRef\]](#)
14. Xia, J.; Wei, J.; Liu, Y.; Zhang, Y.; Guo, S.; Li, C.; Bie, S.; Jiang, J. Design of a Wideband Absorption Frequency Selective Rasorber Based on Double Lossy Layers. *IEEE Trans. Antennas Propag.* **2020**, *68*, 5718–5723. [\[CrossRef\]](#)
15. Sharma, A.; Malik, S.; Ghosh, S.; Srivastava, K.V. A Miniaturized Frequency Selective Rasorber with Independently Regulated Selective Dual-Transmission Response. *IEEE Antennas Wirel. Propag. Lett.* **2022**, *21*, 257–261. [\[CrossRef\]](#)
16. Wang, M.; Xiang, Z.; Li, Y.; Ren, P.; Xu, R.; Gu, C.; Xue, W.; Gao, S. Design of a Dual-Passband Frequency-Selective Rasorber with High Selectivity. *IEEE Antennas Wirel. Propag. Lett.* **2025**, *24*, 2954–2958. [\[CrossRef\]](#)
17. Yu, Q.; Liu, S.; Monorchio, A.; Kong, X.; Brizi, D.; Zhang, X.; Wang, L. Miniaturized Wide-Angle Rasorber with a Wide Interabsorption High Transparent Bandpass Based on Multiple 2.5-D Resonators. *IEEE Antennas Wirel. Propag. Lett.* **2022**, *21*, 416–420. [\[CrossRef\]](#)

18. Gao, C.; Peng, L.; Hu, X.; Zhao, S.; Zhang, X.; Pang, X.; Ye, D. Ultrawideband Rasorber with Quasi-Square Frequency Response Based on Dispersion Engineering. *IEEE Trans. Antennas Propag.* **2026**, *74*, 1665–1678. [[CrossRef](#)]
19. Hao, W.; Zhou, W.; Liu, S.; Liu, J.; Ye, Z.; Geng, X.; Luo, K.; Ji, X. Broadband Frequency-Selective Rasorber Enabled by Offset Resonance for Enhanced Absorption. *IEEE Trans. Microw. Theory Tech.* **2026**, 1–11. [[CrossRef](#)]
20. Chen, Q.; Sang, D.; Guo, M.; Fu, Y. Miniaturized Frequency-Selective Rasorber with a Wide Transmission Band Using Circular Spiral Resonator. *IEEE Trans. Antennas Propag.* **2019**, *67*, 1045–1052. [[CrossRef](#)]
21. Guo, M.; Chen, Q.; Bai, T.; Wei, K.; Fu, Y. Wide Transmission Band Frequency-Selective Rasorber Based on Convolved Resonator. *IEEE Antennas Wirel. Propag. Lett.* **2020**, *19*, 846–850. [[CrossRef](#)]
22. Guo, Q.; Su, J.; Li, Z.; Song, J.; Guan, Y. Miniaturized-Element Frequency-Selective Rasorber Design Using Characteristic Modes Analysis. *IEEE Trans. Antennas Propag.* **2020**, *68*, 6683–6694. [[CrossRef](#)]
23. Yu, Q.; Liu, S.; Monorchio, A.; Kong, X.; Wen, Y.; Huang, Z. A Miniaturized High-Selectivity Frequency Selective Rasorber Based on Subwavelength Resonance and Interdigital Resonator. *IEEE Antennas Wirel. Propag. Lett.* **2019**, *18*, 1833–1837. [[CrossRef](#)]
24. Zhang, H.; Ye, H.; Lu, Z.; Dai, W.; Xiao, M.; Jiang, J.; Miao, L. Miniaturized Lossy-Layer Scheme for Designing a Frequency Selective Rasorber. *IEEE Antennas Wirel. Propag. Lett.* **2022**, *21*, 2472–2476. [[CrossRef](#)]
25. Ge, J.; Jiang, W.; Hong, T.; Gao, Y.; Gong, S. Bandpass Frequency-Selective Rasorber with Wide Transmission Band and High Selectivity Based on Multiorder Resonance. *IEEE Trans. Antennas Propag.* **2023**, *71*, 9621–9632. [[CrossRef](#)]
26. Xu, Y.; Wu, W.; Shi, Q.; Shi, T.; Wang, S. Design of miniaturized and highly selective frequency selective rasorber based on compact spiral resonator. *Electron. Lett.* **2024**, *60*, e70062. [[CrossRef](#)]

**Disclaimer/Publisher’s Note:** The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.